

Your Cancer Summary Report

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Stage IV High-Grade Serous Ovarian Cancer

Pacific Northwest Precision Oncology Center

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⚠️ DRAFT - FOR CLINICIAN REVIEW

This report has not yet been reviewed by your healthcare team. Please wait for your doctor to discuss these results with you.

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At a Glance

Your Diagnosis

Ovarian Cancer - High-Grade Serous Carcinoma (HGSOC)

Stage: Stage IV | **Grade:** High Grade

You have a form of ovarian cancer called high-grade serous carcinoma. It has spread beyond the ovaries to other parts of the body (Stage IV). While this is serious, advanced testing has found several treatment options that may work well for your specific cancer.

✓ Positive Factors

- Your BRCA1 germline mutation makes you eligible for PARP inhibitor therapy (olaparib, niraparib, rucaparib)
- Your tumor's HRD score of 72 confirms your cancer has trouble repairing DNA damage, which means PARP inhibitors are likely to work well
- The tumor shows moderate immune cell activity (antigen presentation score 0.76), suggesting checkpoint inhibitor combinations may help
- PI3K/AKT pathway is activated — Alpelisib (an FDA-approved PI3K inhibitor) may be an effective targeted option
- Your cancer has 20 estimated neoantigens, providing targets for the immune system

⚠ Challenges to Consider

Your cancer is platinum-resistant, which means standard

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- TP53 tumor suppressor loss is present — a common HGSOC feature with no direct targeted therapy yet
- PTEN and CDKN2A deletions contribute to tumor growth and survival

What Your Tests Found

Genetic testing of your tumor revealed the following important findings:

BRCA1

Pathogenic

Pathogenic germline mutation (c.5266dupC, BRCA Exchange Tier 1)

You were born with a change in the BRCA1 gene. This gene normally helps repair DNA damage. Because it is not working properly, your cancer is more sensitive to a type of drug called PARP inhibitors, which prevent cancer cells from repairing their DNA.

Treatment Option: PARP inhibitors FDA-approved: olaparib (Lynparza), niraparib (Zejula), rucaparib (Rubraca)

TP53

Pathogenic

R175H missense mutation (somatic, VAF 73%)

The TP53 gene — your cell's main 'guardian' that stops cancer growth — has a mutation in most cancer cells. This is the most common change seen in high-grade serous ovarian cancer. While there is no pill that directly fixes TP53 yet, it confirms the diagnosis and guides overall treatment strategy.

Treatment Option: No direct FDA-approved therapy; informs overall genomic instability profile

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PIK3CA

Pathogenic

E545K missense mutation (somatic, VAF 42%) + focal amplification (log2=0.72)

The PIK3CA gene — part of a 'growth signal' pathway called PI3K/AKT — is both mutated and amplified (extra copies). This means the pathway is stuck in the 'on' position, helping cancer cells grow. Drugs called PI3K inhibitors may be able to block this.

Treatment Option: Alpelisib (Piqray) FDA-approved for PIK3CA-mutated cancers; upstream regulator analysis Z=9.23

PTEN

Pathogenic

Splice acceptor variant (somatic, VAF 85%) + genomic deletion

PTEN normally acts as a brake on the same PI3K growth pathway. With PTEN lost, that pathway runs unchecked. Together with the PIK3CA mutation, this strongly supports using PI3K/AKT pathway inhibitors.

Treatment Option: Supports PI3K/mTOR pathway inhibition; mTOR inhibitors (everolimus) also relevant

MYC

Pathogenic

Focal amplification (log2 ratio = 1.85, ~3.5x copy gain)

The MYC gene, which tells cells to divide rapidly, has been duplicated several extra times in your cancer. This is a key driver of tumor growth. Spatial analysis confirmed MYC is highly active in the tumor core (log2FC +1.83 vs stroma).

Treatment Option: No direct MYC inhibitor approved; BET bromodomain inhibitors in clinical trials

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CCNE1

Pathogenic

Focal amplification (log2 ratio = 1.12, ~2.2x copy gain)

CCNE1 is the gene for Cyclin E1, which controls when cells start to divide. Extra copies drive uncontrolled cell division in your cancer. GEARS perturbation modeling predicts that blocking CCNE1 would reduce CDK2, CDK4, and MYC activity.

Treatment Option: CDK2 inhibitors (e.g., dinaciclib) in trials; CCNE1 amplification may predict PARP inhibitor sensitivity

CDKN2A

Pathogenic

Homozygous deletion

CDKN2A is another 'cancer brake' gene. When both copies are deleted, cells can divide without limits. This works together with CDK4 amplification to drive rapid tumor growth.

Treatment Option: CDK4/6 inhibitors (palbociclib, ribociclib, abemaciclib) may compensate for CDKN2A loss

Your Tumor Environment

Advanced spatial analysis shows how different cells are organized in your tumor:

Tumor Microenvironment

Spatial transcriptomics analysis of 300 tissue spots across 4 regions (tumor core, stroma, immune zones, necrotic areas) revealed a moderately immune-infiltrated tumor. Tumor cells dominate the core (56 spots), with significant endothelial (44), macrophage (43), fibroblast (41), and mesenchymal (34) populations. CD8+ cytotoxic T-cells were detected in 30 spots, primarily in immune zones adjacent to the tumor core.

Immune Status: • **Mixed immune activity** - Variable immune response across the tumor

We mapped gene activity across your tumor tissue like a detailed geographic map. The most active cancer area (tumor core) is highly proliferating and dominated by cancer cells. Your immune system has T-cells present and active in zones near the tumor, which is a good sign for immunotherapy. However, fibrous tissue (stroma) forms a layer around parts of the tumor that may limit drug delivery. The drug Alpelisib targets growth signals that are very active in your tumor core.

Key Patterns Observed

- Ki67 highly expressed in tumor core (log2FC +1.88 vs stroma, p=0.007), confirming active proliferation
- MYC amplification spatially active in tumor core (log2FC +1.83)
- Stromal markers FAP (-2.03), VIM (-2.36), and ACTA2 (-2.14) enriched outside tumor core — cancer-associated fibroblasts form a barrier
- NNMT downregulated in tumor core (log2FC -1.91) — supports NNMT as metabolic vulnerability in stroma/CAF compartment

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- Spatial autocorrelation (Moran's $I \approx 0$): expression patterns are uniformly distributed, not strongly clustered — may reflect aggressive, infiltrating growth pattern
- Necrotic region represents 74/300 spots (25%), indicating areas of cell death typical of high-grade disease

What the Microscope Shows

The tissue sample viewed under the microscope shows cancer cells with irregular, enlarged nuclei — a hallmark of high-grade disease. About one-quarter of the tissue has areas where cancer cells have died (necrosis), which is common in fast-growing tumors. The highly cellular areas match exactly where the genetic mapping found the most active cancer cells.

Key Observations

- H&E microscopy confirms high-grade serous histology with significant nuclear atypia
- Necrotic regions occupy approximately 25% of tissue section, consistent with Stage IV aggressive disease
- High cellularity zones correspond to tumor core regions identified in spatial transcriptomics
- No evidence of low-grade histological features

Treatment Options

Based on your test results, your care team may consider these treatments:

Olaparib (Lynparza) — PARP Inhibitor

targeted

FDA Approved

Expected Response: 60-70% in BRCA1/2 carriers

Olaparib is a pill that blocks an enzyme called PARP, which cancer cells use to fix their DNA. Because your cancer already has trouble fixing DNA (due to BRCA1 loss and HRD score 72), this drug causes cancer cells to collapse from unrepaired damage — while mostly leaving normal cells alone.

Why this may help you: Your BRCA1 germline mutation and HRD score of 72 (≥ 42 = PARP-eligible) are the strongest predictors of PARP inhibitor benefit. This is the top-priority treatment recommendation.

Common side effects: Fatigue (tiredness), Nausea, Low blood counts, Headache

Alpelisib (Piqray) — PI3K Inhibitor

targeted

ESCAT I

Expected Response: 35-40% in PIK3CA-mutated tumors

Alpelisib is a pill that blocks the PI3K enzyme, which is stuck in the 'on' position due to your PIK3CA mutation. Turning this signal off can slow tumor growth. It is often combined with another drug called fulvestrant.

Why this may help you: PIK3CA E545K mutation plus PTEN deletion plus spatial DE analysis all point to active PI3K signaling. Upstream regulator analysis identified Alpelisib as the top drug recommendation (Z-score = 9.23).

Common side effects: High blood sugar (requires monitoring), Diarrhea

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Pembrolizumab (Keytruda) — Checkpoint Inhibitor

immunotherapy

ESCAT II

Expected Response: 10-20% in platinum-resistant HGSOC

Pembrolizumab helps your immune system recognize and attack cancer cells. It works by releasing a 'brake' on T cells so they can fight the tumor. Spatial analysis showed moderate immune infiltration in your tumor, suggesting the immune system can be activated.

Why this may help you: Antigen presentation pathway score of 0.76 with intact HLA, good B2M and TAP1/TAP2 expression. 20 estimated neoantigens provide immune targets. CD8+ T-cells present in 30 spatial spots. Consider combination with PARP inhibitor.

Common side effects: Fatigue, Immune-related reactions (lung, liver, gut), Rash, Thyroid problems

CDK4/6 Inhibitor (e.g., Palbociclib) — Clinical Trial

targeted

Clinical Trial Only

Expected Response: Under investigation in HGSOC

CDK4/6 inhibitors are pills that slow cancer cell division by blocking two enzymes (CDK4 and CDK6). Because your cancer has extra copies of CDK4 and has lost its natural brakes (CDKN2A deleted), these drugs may be particularly effective.

Why this may help you: CDK4 amplification ($\log_2=0.65$) combined with CDKN2A homozygous deletion and GEARS GNN prediction of CDK4 downregulation upon CCNE1 knockdown all support CDK pathway targeting.

Common side effects: Low blood counts (neutropenia), Fatigue, Nausea,

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Liposomal Doxorubicin (Doxil) — Chemotherapy

chemotherapy

NCCN Category 1

Expected Response: 15-26% in platinum-resistant HGSOC

Doxil is a chemotherapy drug packaged in tiny fat bubbles that help it reach tumor tissue. It damages cancer cell DNA. While your cancer is resistant to platinum drugs, Doxil works differently and remains a standard option for platinum-resistant ovarian cancer.

Why this may help you: Standard-of-care option for platinum-resistant HGSOC. Remains effective independently of BRCA status.

Common side effects: Hand-foot skin reaction, Mouth sores, Fatigue, Nausea

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Clinical Trials

You may be eligible for these research studies offering new treatments:

SORAYA: Mirvetuximab Soravtansine in FRα-High Platinum-Resistant Ovarian Cancer

Phase: 3 | **Status:** Active, not recruiting

Trial ID: NCT03574779

This trial tested a drug (mirvetuximab soravtansine/ELAHERE) that targets a protein called FRα, which is often high in HGSOC. The drug recently received FDA accelerated approval for platinum-resistant ovarian cancer.

Nearest Location: Multiple US sites

Contact: [clinicaltrials.gov NCT03574779](https://clinicaltrials.gov/NCT03574779)

PARP + CDK4/6 Inhibitor Combination in BRCA-mutated Solid Tumors

Phase: 2 | **Status:** Recruiting

Trial ID: NCT04736953

Combines olaparib (PARP inhibitor matching your BRCA1 mutation) with a CDK4/6 inhibitor (matching your CDK4 amplification and CDKN2A deletion). Nearby site in Seattle.

Nearest Location: Fred Hutchinson Cancer Center, Seattle WA

Contact: Contact Fred Hutchinson Cancer Center Oncology Research: 206-667-4355

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PICCASSO: PI3K Inhibitor + Checkpoint Inhibitor in Platinum-Resistant Ovarian Cancer

Phase: 2 | **Status:** Recruiting

Trial ID: NCT05009056

Combines a PI3K inhibitor (targeting your PIK3CA mutation) with pembrolizumab (checkpoint inhibitor matching your moderate immune score). Directly relevant to your tumor biology.

Nearest Location: Seattle Cancer Care Alliance

Contact: Contact SCCA: 206-288-7222

Your Monitoring Plan

Regular check-ups help us track your health and catch any changes early.

Scheduled Tests

Every 6-8 weeks during treatment	CA-125 blood test Track tumor marker response to treatment; significant drops indicate the cancer is responding
Every 3 months (or after 2-3 cycles of treatment)	CT scan of chest, abdomen, and pelvis Measure physical tumor size and check for new areas of spread
Every 3-4 weeks during treatment	Complete blood count (CBC) and metabolic panel Check for low blood counts (anemia, neutropenia), kidney function, and blood sugar (especially if on Alpelisib)
Before starting and every 6 months during doxorubicin therapy	Echocardiogram Monitor heart function if liposomal doxorubicin is used
Once (if not yet completed)	Germline genetic counseling session Discuss BRCA1 implications for you and your family members with a certified genetic counselor

When to Call Your Doctor

Contact your care team right away if you experience:

- Sudden or worsening abdominal pain or bloating
- Shortness of breath or new chest pain
- Fever above 100.4°F (38°C) — may indicate infection from low blood counts
- Unusual bleeding or bruising

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- New or worsening hand-foot redness, swelling, or blistering
- Signs of allergic reaction: hives, facial swelling, difficulty swallowing

Questions? Your oncology nurse navigator or on-call oncologist. After hours: Pacific Northwest Precision Oncology Center 24-hour line.

For Your Family

Your BRCA1 mutation was present from birth (germline), meaning it was inherited and can be passed to biological children. First-degree relatives (parents, siblings, children) have a 50% chance of carrying the same mutation. This mutation also increases risk for breast cancer (up to 72% lifetime), pancreatic cancer, and in males, prostate cancer.

Recommended Steps for Family Members

- Discuss BRCA1 test results with your first-degree relatives (parents, siblings, children)
- Daughters and sons should consider BRCA1 genetic testing starting at age 25
- Sisters and daughters who test positive should consider increased breast/ovarian screening and risk-reduction options
- Share results with your primary care physician who can coordinate family referrals

Genetic Counseling Recommended: A genetic counselor can help explain what these results mean for your family members and discuss testing options.

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Resources & Support

You don't have to face this alone. These resources can help:



National Ovarian Cancer Coalition (NOCC)

Provides education, support groups, and financial assistance navigation for ovarian cancer patients and families.

📞 1-888-682-7426 | 🌐 <https://www.ovarian.org>



FORCE: Facing Our Risk of Cancer Empowered

Specializes in hereditary breast/ovarian cancer (BRCA mutations). Offers peer support, genetic counseling navigation, and information on BRCA-specific clinical trials.

📞 1-866-288-7475 | 🌐 <https://www.facingourrisk.org>



Patient Advocate Foundation

Helps navigate insurance appeals, co-pay assistance, and financial toxicity. Can help with olaparib (Lynparza) and alpelisib (Piqray) co-pay programs.

📞 1-800-532-5274 | 🌐 <https://www.patientadvocate.org>



Ovarian Cancer Research Alliance (OCRA)

Connects patients with local and online support groups, peer mentors who are HGSOC survivors, and the latest clinical trial listings.

| 🌐 <https://ocrahope.org>

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Data Sources: mcp-epic (clinical/lab data), mcp-genomic-results (somatic VCF + CNV), mcp-multiomics (PDX RNA/protein/phospho integration), mcp-spatialtools (spatial transcriptomics 300 spots), mcp-openimagedata (H&E histology), mcp-neoantigen (neoantigen burden + HLA scoring), mcp-perturbation (GEARS GNN perturbation modeling), mcp-opentargets (target tractability), mcp-clinicaltrials (trial matching)

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